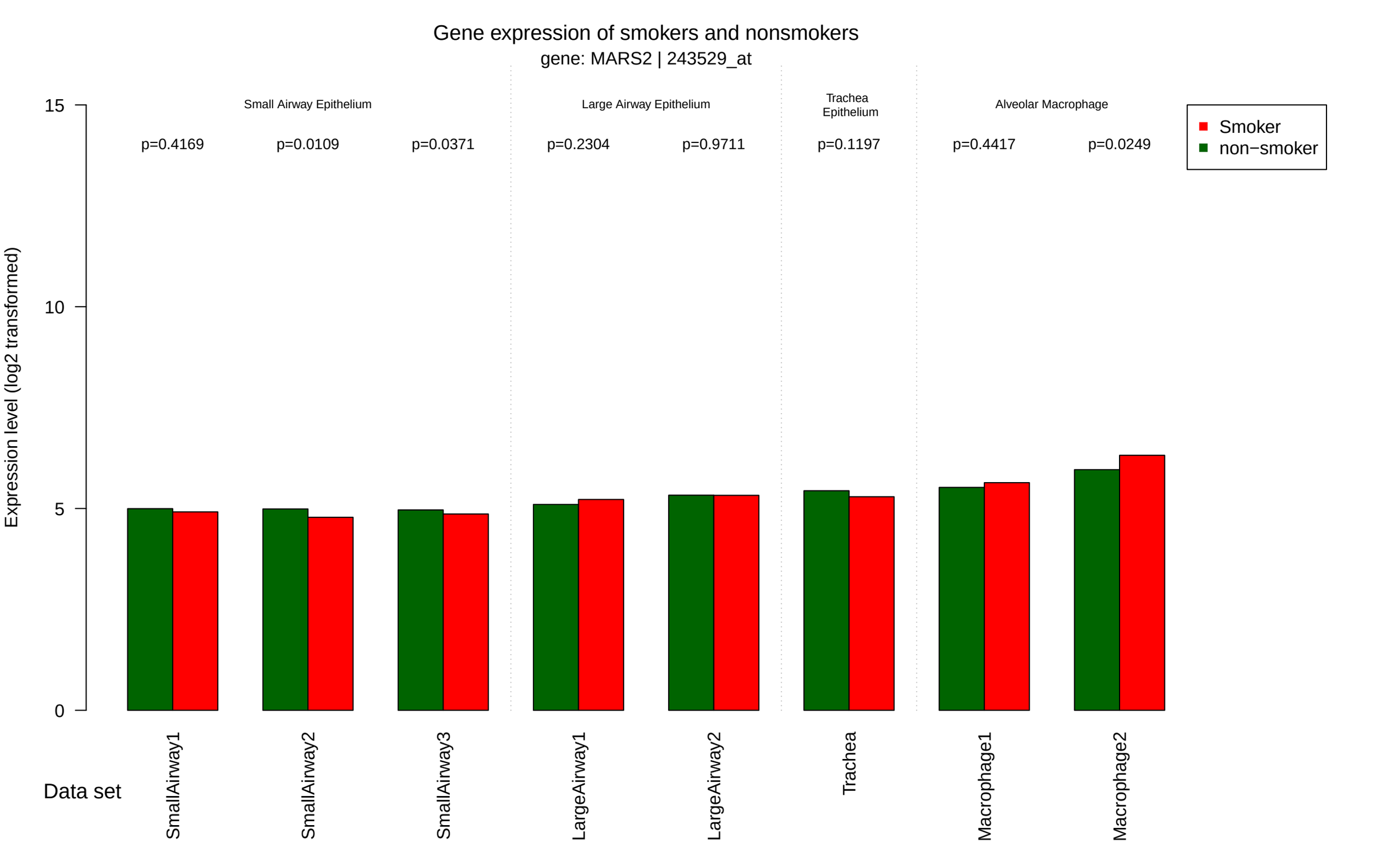
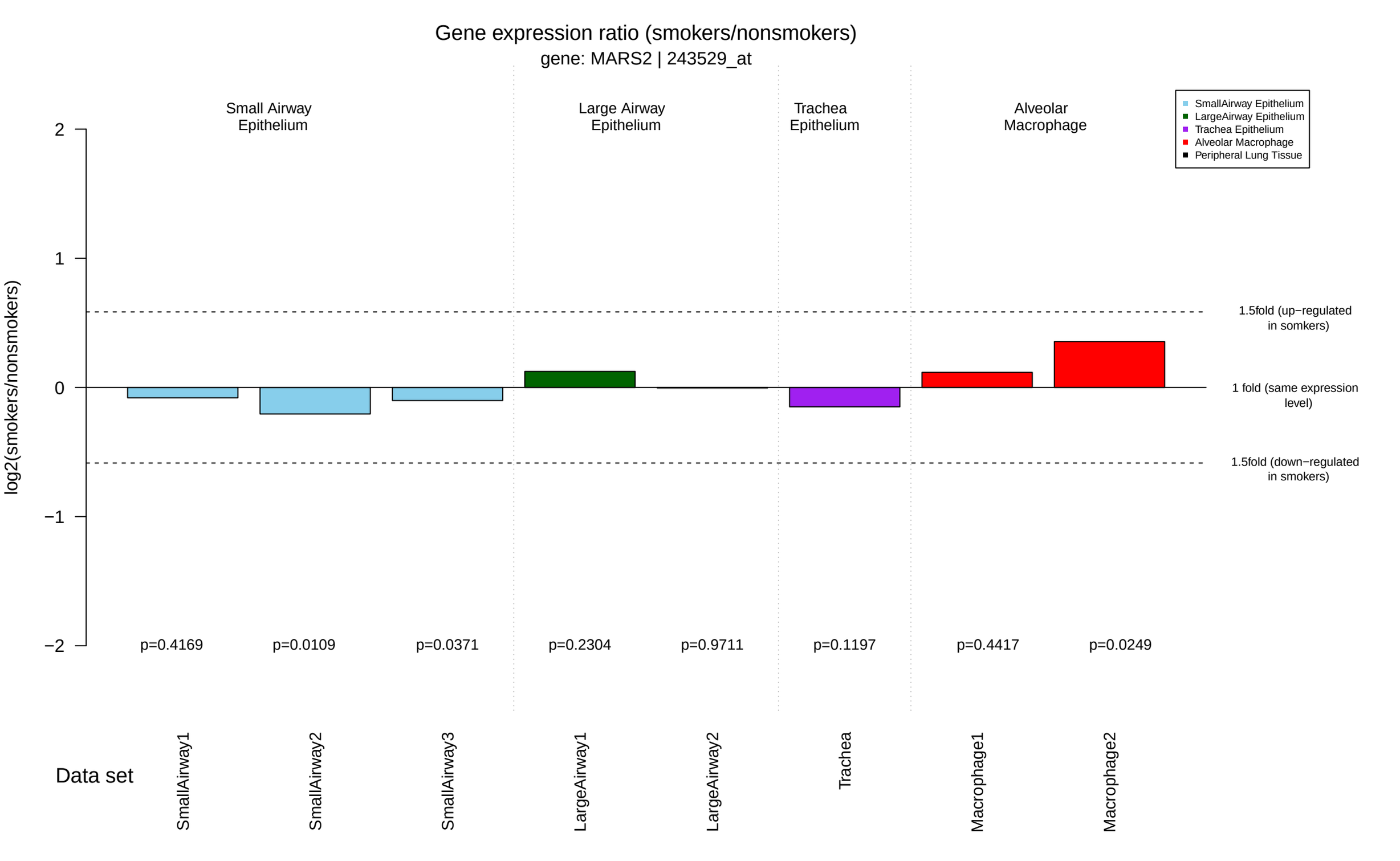
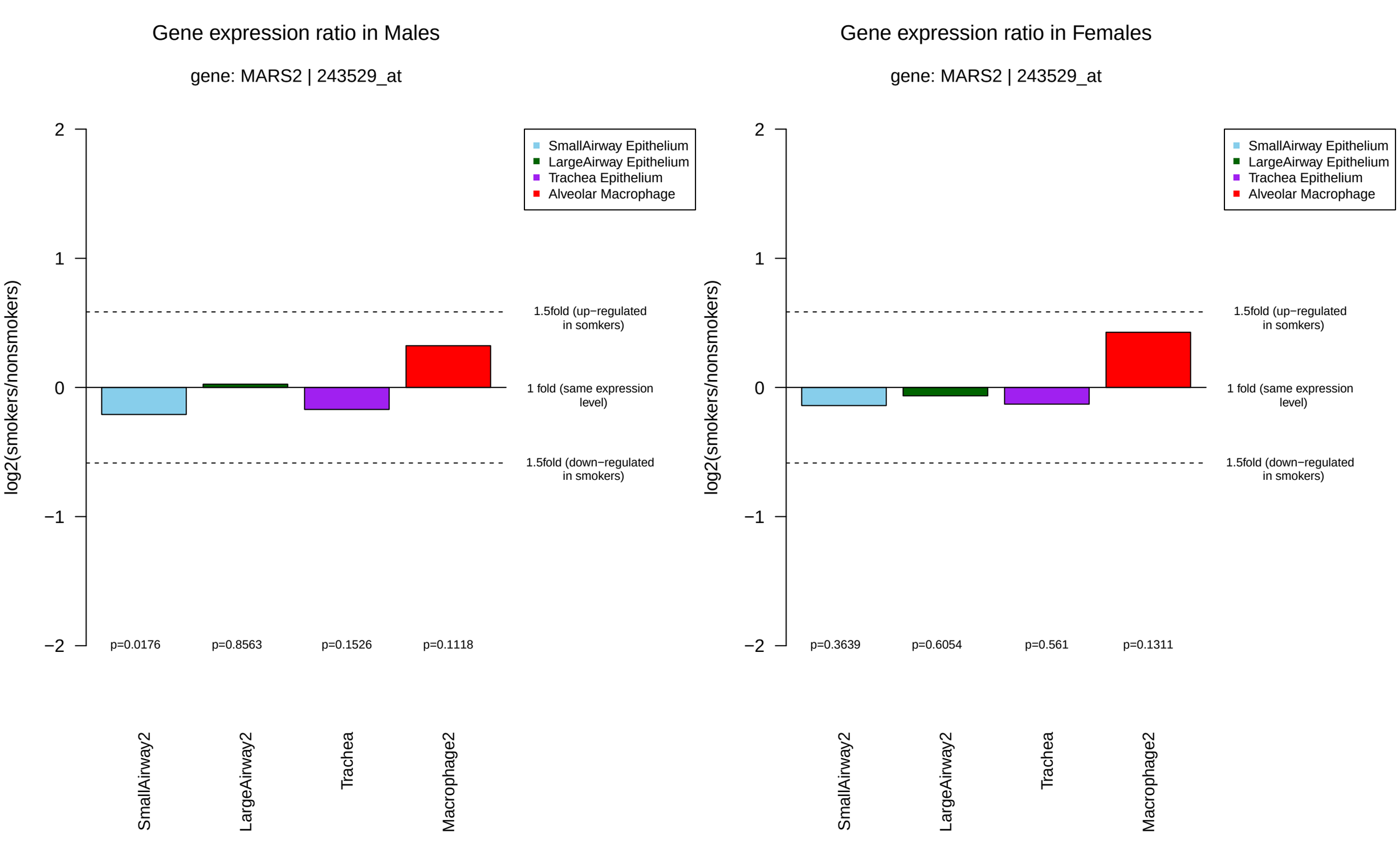
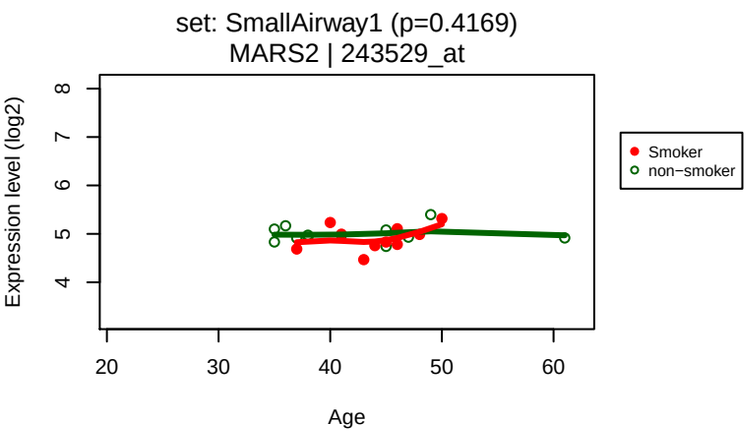
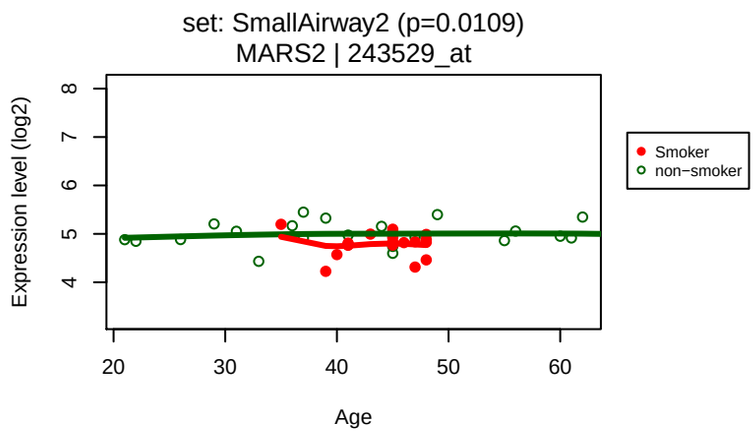
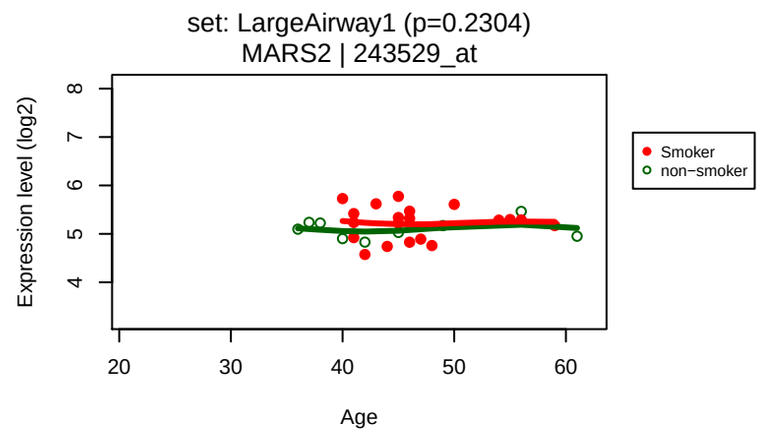

Gene expression of smokers and nonsmokers
gene: MARS2 | 243529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4169
p=0.0109
p=0.0371
p=0.2304
p=0.9711
p=0.1197
p=0.4417
p=0.0249
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

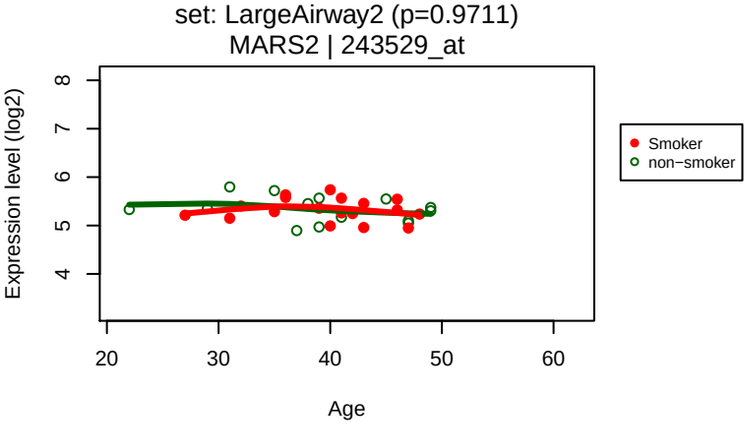

Gene expression ratio (smokers/nonsmokers)
gene: MARS2 | 243529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4169
p=0.0109
p=0.0371
p=0.2304
p=0.9711
p=0.1197
p=0.4417
p=0.0249
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

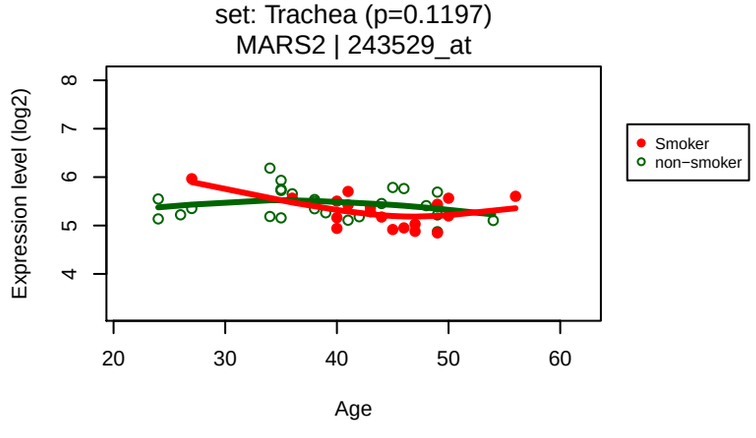

Gene expression ratio in Males
gene: MARS2 | 243529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0176
p=0.8563
p=0.1526
p=0.1118
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MARS2 | 243529_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3639
p=0.6054
p=0.561
p=0.1311
SmallAirway2
LargeAirway2
Trachea
Macrophage2

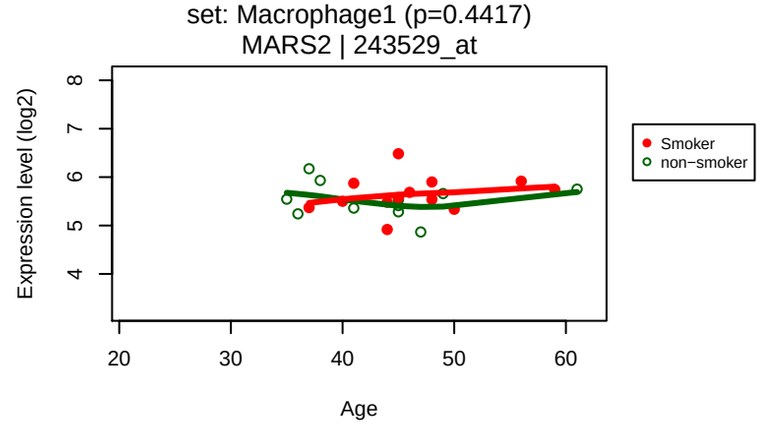

set: SmallAirway1 (p=0.4169)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker

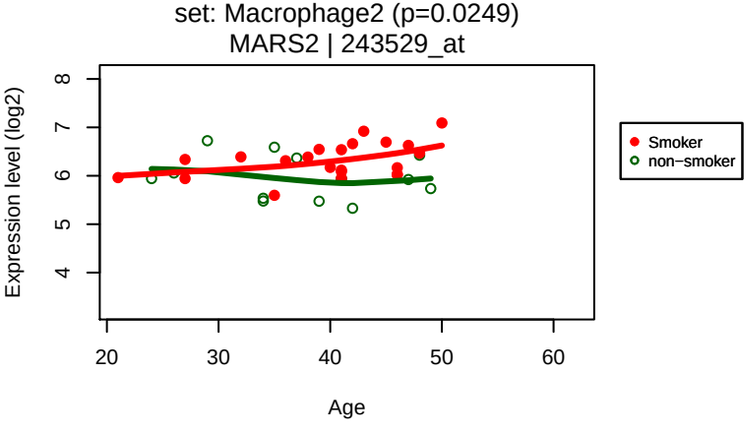

set: SmallAirway2 (p=0.0109)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker

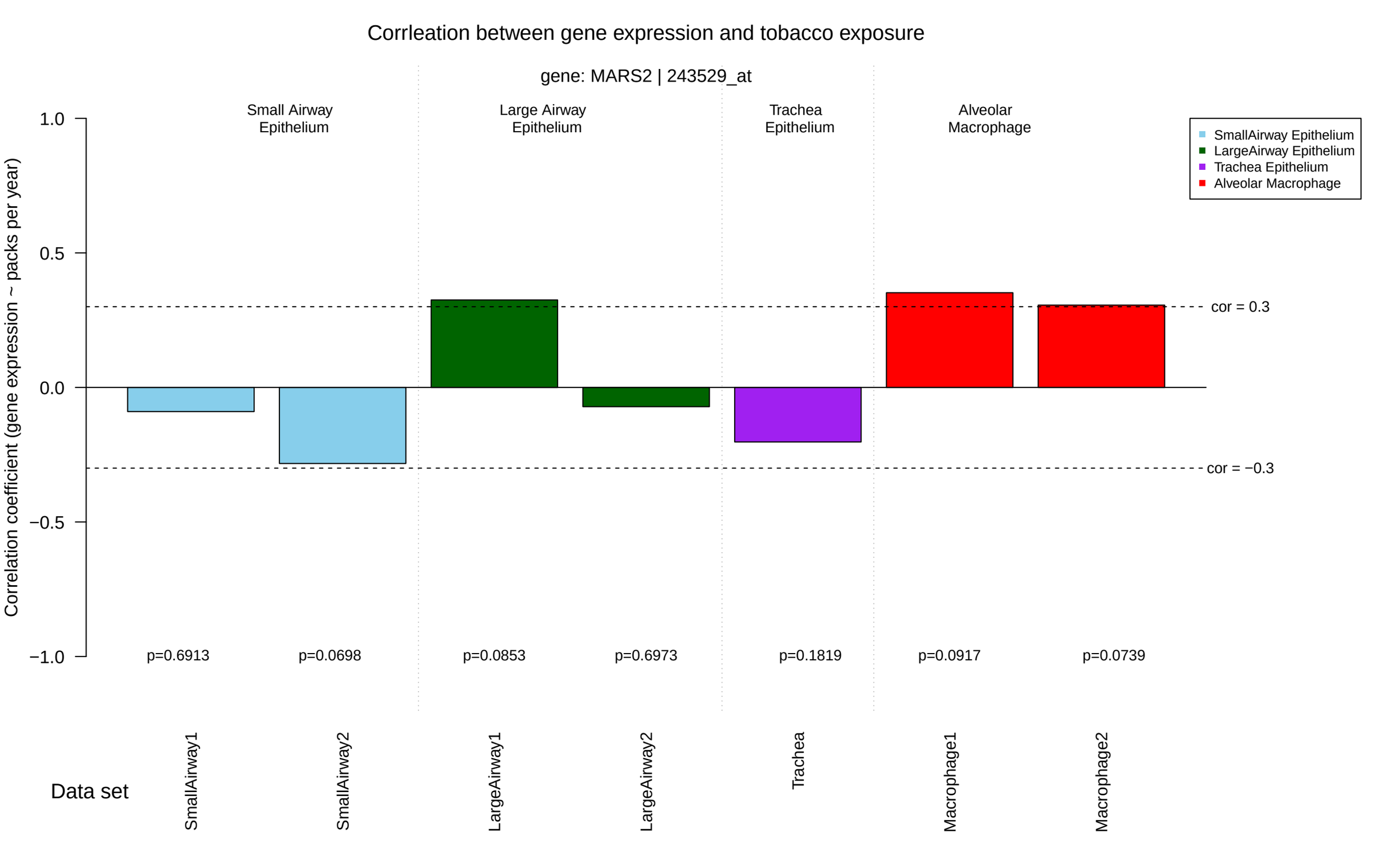

set: LargeAirway1 (p=0.2304)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9711)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1197)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4417)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0249)
MARS2 | 243529_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MARS2 | 243529_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6913
p=0.0698
p=0.0853
p=0.6973
p=0.1819
p=0.0917
p=0.0739
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2